

multiomicsGEP — Project Status

Supervised Bayesian Matrix Factorization for Joint Genomics + Survival Modelling

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0.1 Project Overview

This project implements a **Supervised Bayesian Matrix Factorization** model that jointly decomposes high-dimensional genomics data (gene expression, DNA methylation, etc.) and models patient survival outcomes. The core idea is to find a shared latent space that simultaneously explains genomic variation *and* predicts clinical outcomes.

The model:

- **Genomics:** $Y = L F' + E$, where L ($n \times K$) are patient loadings, F ($p \times K$) are biological factor weights, and E is Gaussian noise with feature-specific precision τ_{ij} .
- **Survival:** Cox proportional hazards $h(t_i) = h_0(t_i) \exp(\sum_k l_{ik} * \beta_k)$, linking the same latent loadings L to time-to-event data via survival coefficients β .

Inference: Coordinate Ascent Variational Inference (CAVI), where each parameter update reduces to an Empirical Bayes Normal Means (EBNM) problem with point-normal priors.

0.2 Repository Map

```

multiomicsGEP/
|-- CLAUDE.md                <- Claude Code entry point (thin; defers here)
|-- PROJECT_STATUS.qmd/.pdf  <- This file (project documentation)
|-- README.md                <- GitHub front page
|-- code/
|   |-- SupervisedMF_Context.md <- AI/collaborator quick-reference for V2 code
|   |-- legacy/Supervised_Bayesian_MF.R <- V1: Original CAVI implementation (archived)
|   |-- Supervised_Bayesian_MF_V2.R <- V2: Corrected implementation (all fixes) -- USE THIS
|   |-- update_beta.R          <- Modular beta update functions
|   |-- update_L.R             <- Modular q(L) update functions (vector EBNM, dual-source)
|   |-- update_F.R             <- Modular q(F) update functions (pure genomics, tau cancella
|   |-- update_tau.R           <- Modular q(tau) update functions (closed-form MLE)
|   `-- multiomicsGEP_code.Rmd <- Earlier exploratory R Markdown notebook
|-- derivations/
|   |-- qB/
|   |   |-- qBeta_update_derivation.tex <- Full step-by-step q(beta) derivation (11 pages)
|   |   `-- qBeta_update_derivation.pdf
|   |-- qL/
|   |   |-- qL_update_derivation.tex <- Vector EBNM, dual-source (genomics+survival)

```

```

| | `-- qL_update_derivation.pdf
| |-- qF/
| | |-- qF_update_derivation.tex <- tau cancellation property, pure genomics
| | `-- qF_update_derivation.pdf
| |-- qTau/
| | |-- qTau_update_derivation.tex <- Closed-form MLE, variance correction
| | `-- qTau_update_derivation.pdf
| |-- EBMF/
| | `-- EBMF_Derivations*.pdf <- Empirical Bayes MF theory notes
| |-- MF_UpdateDerivations/
| | |-- MF_Derivations_UpdateAlgo_2_12_26.pdf <- Reviewed draft -- HAS ERRORS (R1-R8)
| | |-- MF_Derivations_UpdateAlgo_REVISED.tex/.pdf <- Corrected (21 pages)
| | `-- MF_V2_Companion.tex/.pdf <- Math <-> code mapping (17 pages)
| `-- SurvivalMF/
|   `-- *.pdf <- Survival + MF background notes
|-- results/
| |-- run_simulation.R <- Reproduce V2 simulation (8 figs, 11 CSVs)
| |-- run_modular_simulation.R <- Reproduce modular simulation (standalone)
| |-- simulation_report.qmd/.pdf <- V2 simulation report (Quarto, ggplot2)
| |-- modular_sim_report.qmd/.pdf <- Modular simulation report (Quarto, per-fig commentar
| |-- simulation_console_output.txt
| |-- figures/
| | |-- full_sim/ <- 8 figure pairs from V2 simulation
| | `-- modular_sim/ <- 8 figure pairs from modular simulation
| `-- tables/
|   |-- full_sim/ <- 11 CSV tables from V2 simulation
|   `-- modular_sim/ <- 11 CSV tables from modular simulation
|-- tests/
| |-- test_helpers.R <- Lightweight assertion framework
| |-- test_update_beta.R <- 24 tests for update_beta.R (9 groups)
| |-- test_update_L.R <- 28 tests for update_L.R (9 groups)
| |-- test_update_F.R <- 26 tests for update_F.R (9 groups)
| |-- test_update_tau.R <- 27 tests for update_tau.R (9 groups)
| `-- run_tests.R <- Master runner: 105/105 tests passing
|-- demos/
| |-- demo_update_beta.R <- 5 demos for update_beta.R
| |-- demo_update_L.R <- 5 demos for update_L.R
| |-- demo_update_F.R <- 5 demos for update_F.R
| `-- demo_update_tau.R <- 5 demos for update_tau.R
|-- docs/
| |-- Makefile <- `make all` renders .qmd -> .pdf + .html via Quarto
| |-- update_beta.qmd/.pdf/.html <- Companion doc for update_beta.R
| |-- update_L.qmd/.pdf/.html <- Companion doc for update_L.R
| |-- update_F.qmd/.pdf/.html <- Companion doc for update_F.R
| `-- update_tau.qmd/.pdf/.html <- Companion doc for update_tau.R
`-- paper/
  `-- multiomicsGEP_manuscript.qmd <- Manuscript draft (Quarto)

```

0.3 Mathematical Framework

0.3.1 Model Components

Symbol	Dimension	Meaning
Y	$n \times p$	Observed genomics data matrix
L	$n \times K$	Patient loading matrix (shared latent space)
F	$p \times K$	Biological factor weight matrix
β	$K \times 1$	Survival coefficient vector
τ_j	p-vector	Feature-specific noise precision
(t_i, δ_i)	n-vectors	Survival time and event indicator

0.3.2 Inference Strategy

1. **Mean-field factorisation:** $q(L, F, \beta) = q(L) q(F) q(\beta)$
2. **CAVI loop:** For each factor $k = 1, \dots, K$:
 - Update $q(l_k)$ via EBNM($x = B_L/A_L$, $s = 1/\sqrt{A_L}$)
 - Update $q(f_k)$ via EBNM($x = B_F/A_F$, $s = 1/\sqrt{A_F}$)
 - Update $q(\beta_k)$ via EBNM($x = B_\beta/A_\beta$, $s = 1/\sqrt{A_\beta}$)
3. **Update τ_j** from expected squared residuals
4. **Cox Taylor expansion** linearises the non-conjugate survival likelihood into a weighted Gaussian form with working response z_i and weight W_{ii}

0.3.3 Key Mathematical Concepts

- **Posterior means vs second moments:** $EL = E_q[l]$, $EL2 = E_q[l^2] = \text{Var}_q(l) + \text{mean}^2$. This distinction is critical for the variance inflation term in the τ update and the error-in-variables correction in the β update.
- **EBNM sub-problems:** Each coordinate update reduces to: given pseudo-observations x with noise s , estimate a prior g and posterior q .
- **Error-in-variables:** Using $E[l^2]$ (not $\bar{l}^2$) in the β precision prevents overfitting to uncertain loadings.

0.4 Derivation Review Summary

A thorough review of MF_Derivations_UpdateAlgo_2_12_26.pdf identified **8 errors**, all corrected in MF_Derivations_UpdateAlgo_REVISIED.tex.

⚠ Derivation errors in 2_12_26.pdf (R1–R8)

The February 12 draft contains 8 errors listed below. Use **REVISED.pdf** for all reference.

ID	Location	Error	Correction
R1	Sec. 1A	Wrong dimension subscripts on Y , L , F , E	Fixed: $Y_{(n \times p)} = L_{(n \times K)} F'_{(K \times p)} + E_{(n \times p)}$
R2	Eq. 22	$z^{\{-k\}i}$ <i>defined with $f_{\{k'\}}$</i> instead of $\beta_{\{k'\}}$	Fixed: uses $\beta_{\{k'\}}$ (survival coefficient)
R3	Eqs. 32-35	$f^2_{\{jk\}}$, β^2_k where E_q second moments needed	Added explicit E_q notation
R4	Eqs. 47-48	EBNM inputs subscripted i (sample)	Fixed to j (feature) for F update
R5	Eq. 62	Missing exponent: β_k should be β^2_k	Added squared exponent
R6	Sec. 6	β update as generic WLS; no EBNM form	Derived full A_k , B_k for EBNM
R7	Eq. 83	Sign error: $\log \tau + \tau * R\text{-bar}^2$	Fixed to $\log \tau - \tau * R\text{-bar}^2$
R8	Eq. 85	$l^2_{\{jk\}}$ wrong index (j should be i)	Fixed to $l^2_{\{ik\}}$

0.4.1 Document Relationships

- **2_12_26.pdf** — reviewed document; contains errors R1–R8
- **REVISED.tex/.pdf** — corrected version with all 8 fixes + errata table (21 pages)
- **MF_V2_Companion.tex/.pdf** — comprehensive companion mapping math to V2 code (17 pages)

0.5 Code Versions

0.5.1 V1: code/legacy/Supervised_Bayesian_MF.R

The original implementation. Functionally correct but with several known issues:

- Orthogonalisation always ON; resets EL2/EF2 to squared means (drops posterior variance every 10 iterations)
- `z_no_k` computed from start-of-iteration eta only (not true Gauss-Seidel)
- No numerical floors on EBNM precision inputs
- Convergence checks only delta-L (not delta-beta)
- Only RMSE tracked (no ELBO monitoring)

0.5.2 V2: Supervised_Bayesian_MF_V2.R

Recommended implementation

Use `code/Supervised_Bayesian_MF_V2.R` for all runs. V1 is retained for reference only.

The corrected implementation with 6 algorithmic improvements:

ID	Change	Description
A1	Gauss-Seidel CAVI	<code>z_no_k</code> recomputed from current EL, EBeta inside k-loop
A2	Orthogonalisation flag	Behind <code>orthogonalize=FALSE</code> (default OFF)
A3	Precision floors	<code>pmax(..., 1e-10)</code> on <code>A_L</code> , <code>A_F</code> , <code>A_Beta</code>
A4	Dual convergence	Both mean
A5	ELBO proxy	Genomics log-likelihood tracked per iteration
A6	Taylor refresh flag	<code>refresh_taylor=FALSE</code> for optional per-k recomputation

Function signature:

```
fit_supervised_mf(Y, time, status, K=5, max_iter=100, tol=1e-5,
                  orthogonalize=FALSE, refresh_taylor=FALSE, verbose=TRUE)
# Returns: list(L, F, Beta, Beta2, Tau, history)
```

0.6 How to Run

0.6.1 Prerequisites

```
install.packages(c("survival", "ebnm"))
```

0.6.2 Running the Simulations

V2 (monolithic):

```
source("code/Supervised_Bayesian_MF_V2.R")
```

Modular (standalone, recommended for understanding the algorithm):

```
Rscript results/run_modular_simulation.R
```

Both use the same parameters ($n=250$, $p=1000$, $K=5$, $\text{seed}=42$) and produce equivalent results. The modular script sources only the four standalone update modules; it does not depend on V2.R.

0.6.3 Expected Output

- **RMSE** converges near 1.0 (the true noise SD)
- **ELBO proxy** is non-decreasing
- **Estimated beta** recovers sign pattern (+, -, +, -, 0)
- **Supervised C-index** ~0.86 on held-out tertiles
- **Censoring rate** ~30-40%

0.6.4 Compiling LaTeX Documents

```
cd derivations/MF_UpdateDerivations/  
pdflatex MF_Derivations_UpdateAlgo_REVISED.tex  
pdflatex MF_V2_Companion.tex  
pdflatex MF_V2_Companion.tex    # second pass for cross-references  
  
cd derivations/qB/  
pdflatex qBeta_update_derivation.tex    # run 3x for stable bookmarks
```

0.6.5 Running the Test Suite

i Test suite status: 105/105 passing

Run after any change to a modular update script: `Rscript tests/run_tests.R`

```
Rscript tests/run_tests.R  
# Expected: 105/105 tests passed (24 beta + 28 L + 26 F + 27 tau)
```

0.6.6 Running the Modular Update Demos

```
Rscript demos/demo_update_beta.R      # beta update: 5 demos
Rscript demos/demo_update_L.R          # q(L) update: 5 demos
Rscript demos/demo_update_F.R          # q(F) update: 5 demos -- see tau cancellation
Rscript demos/demo_update_tau.R        # q(tau) update: 5 demos -- see variance correction
```

0.7 Current Status & Next Steps

0.7.1 Completed

- ☒ Derivation review: 8 errors (R1-R8) identified and corrected
- ☒ REVISED.tex/PDF: Corrected derivation document with errata table (21 pages)
- ☒ REVISED.tex expanded: Full step-by-step algebra added throughout (Taylor, L/F/beta/tau updates) with new `derivbox` gray-box environment for intermediate steps
- ☒ V2.R: Corrected R implementation with all 6 algorithmic fixes (A1-A6)
- ☒ V2_Companion.tex/PDF: Comprehensive math-to-code companion document (17 pages)
- ☒ PROJECT_STATUS + SupervisedMF_Context.md: Project documentation
- ☒ Simulation validation: V2 runs end-to-end on synthetic data ($n=250$, $p=1000$, $K=5$)
- ☒ results/: Full simulation outputs – 8 figures, 11 CSV tables, console log
- ☒ derivations/qB/qBeta_update_derivation.tex/.pdf: Self-contained $q(\text{beta})$ derivation (11 pages)
- ☒ code/update_beta.R: Modular beta update (`compute_z_no_k`, `update_beta_k`, `update_beta_all`)
- ☒ tests/: Test infrastructure + 24 tests for beta update (9 groups, all passing)
- ☒ demos/demo_update_beta.R: 5 interactive demos for beta update
- ☒ code/update_L.R: Modular $q(L)$ update – vector EBNM, dual-source
- ☒ code/update_F.R: Modular $q(F)$ update – pure genomics, tau cancellation
- ☒ code/update_tau.R: Modular $q(\text{tau})$ update – closed-form MLE
- ☒ tests/test_update_L.R: 28 tests (9 groups, all passing)
- ☒ tests/test_update_F.R: 26 tests (9 groups, tau cancellation verified)
- ☒ tests/test_update_tau.R: 27 tests (9 groups, variance correction verified)
- ☒ tests/run_tests.R: Updated master runner – 105/105 tests passing
- ☒ demos/demo_update_L.R: 5 demos (anatomy, signal recovery, genomics vs combined, error-in-variables, $K=5$)
- ☒ demos/demo_update_F.R: 5 demos (tau cancellation, sparse recovery, differential shrinkage, sparsity, $K=5$)
- ☒ demos/demo_update_tau.R: 5 demos (anatomy, known noise, variance correction, heteroscedastic, ELBO proxy)
- ☒ derivations/qL/qL_update_derivation.tex/.pdf: Self-contained $q(L)$ derivation
- ☒ derivations/qF/qF_update_derivation.tex/.pdf: Self-contained $q(F)$ derivation
- ☒ derivations/qTau/qTau_update_derivation.tex/.pdf: Self-contained $q(\text{tau})$ derivation
- ☒ docs/: Companion documentation for all 4 modular update scripts (.md + .pdf + .html)

- ☒ roxygen2 @export/@family/@seealso tags added to all 11 functions
- ☒ Targeted inline comments added to all 4 R scripts
- ☒ results/ reorganized: figures/{full_sim,modular_sim}/ and tables/{full_sim,modular_sim}/
- ☒ results/run_modular_simulation.R: Standalone CAVI script sourcing only the 4 modular modules
- ☒ results/modular_sim_report.qmd/.pdf: Quarto simulation report for modular implementation
- ☒ results/simulation_report.qmd/.pdf: Quarto simulation report for V2 implementation
- ☒ CLAUDE.md created; PROJECT_STATUS converted to .qmd for PDF rendering

0.7.2 Potential Next Steps

Note

List maintenance: Mark items [x] when implemented and confirmed, or explicitly dropped. Do not remove them — checked items are a record of accomplishments. Only remove true duplicates or items added in error.

Near-term (next 1–2 sessions):

- ☐ **Hold-out prediction task:** Train on 80% of each cohort, evaluate C-index on 20% held-out. Requires `predict_supervised_mf()`: fix learned F and β , run a single EBNM pass on Y_{test} to recover L_{test} , compute $L_{\text{test}} \%*\% EBeta$ as linear predictor. Highest-priority performance validation.
- ☐ **K selection:** $K=5$ is fixed across all cohorts; likely suboptimal for small- n cohorts (Dijk $n=90$, PACA_AU_seq $n=52$). Options, roughly by cost:
 1. *Fit large K and prune* – fit $K=10$, let EBNM zero uninformative factors; count surviving non-zero factors as effective K . Free at inference time.
 2. *Scree plot / PVE elbow* – plot per-factor explained variance from existing `factor_summary_table.csv` identify natural elbow. Zero additional compute.
 3. *Held-out C-index CV* – k -fold over candidate K values; computationally intensive, best run on Longleaf HPC.
- ☐ **Lab meeting slide deck (April 9, 2026):** Convert PDAC report to a Quarto reveals presentation covering: model motivation, CAVI overview, per-dataset results, cross-dataset summary, pooled fit, and future directions.

Medium-term:

- ☐ **Survival-aware feature selection:** Replace top- N -by-variance with a univariate Cox pre-filter or survival-weighted variance metric, to bias the input feature set toward prognostically relevant genes before factorization.
- ☐ **Batch correction before pooling:** Apply `ComBat` or `limma::removeBatchEffect` before `pool_datasets()` to reduce dataset-of-origin effects in the pooled RNA-seq model.

- **Prior family exploration:** Test `prior_family = "point_laplace" or "normal_scale_mixture"` in EBNM calls; may recover denser or more graded factor structures on real data vs. the default point-normal.
- **Full ELBO tracking:** Currently only the genomics log-likelihood proxy is tracked. Including the survival Taylor term and all KL divergences enables proper model selection and tighter convergence monitoring.
- **Multiple random initializations:** SVD initialization is deterministic. Running from multiple random starts and selecting the best ELBO would reduce sensitivity to local optima, at the cost of $K \times$ compute.

Longer-term:

- **Multi-dataset shared factor space:** Shared F and β with per-cohort L and τ . Requires new update equations and extension of `fit_modular.R`.
- **Multi-modal integration:** Each modality (RNA-seq, proteomics, methylation) gets its own $F^{(m)}$ and $\tau^{(m)}$ sharing a common L . Requires new `update_F_{<m>.R` and extended `update_L.R` accumulating signal across modalities.
- **Manuscript:** Complete paper/`multiomicsGEP_manuscript.qmd` once prediction results and K selection are in hand.

0.8 Session Log

0.8.1 Session 1 (March 4–5, 2026)

Derivation review and code correction session.

1. Reviewed `MF_Derivations_UpdateAlgo_2_12_26.pdf` against the code
2. Identified 8 derivation errors (R1–R8)
3. Created `MF_Derivations_UpdateAlgo_REVISED.tex` with all corrections + errata table
4. Analysed V1 code against corrected derivations; identified 6 improvements
5. Created `Supervised_Bayesian_MF_V2.R` with all algorithmic fixes (A1–A6)
6. Created `MF_V2_Companion.tex` (math to code companion, 17 pages)
7. Created `PROJECT_STATUS.md` and `SupervisedMF_Context.md`

Key decisions:

- Orthogonalisation defaults to OFF in V2 (Point-Normal EBNM promotes sparsity/distinctness without rotation)
- Taylor refresh defaults to OFF (standard IRLS-within-VI; more stable)
- V1 preserved as-is for reference; V2 is the recommended implementation

Deliverables committed: pushed to `origin/main`

0.8.2 Session 2 (March 5, 2026)

Simulation validation, derivation expansion, and documentation session.

1. **Expanded REVISED.tex** from ~650 lines to 1,199 lines with full intermediate algebra throughout every update section:
 - Added new `derivbox` `tcolorbox` (gray) for algebra steps, distinct from `correctionbox` (red) and `ebnmbox` (blue)
 - **Taylor (Sec 3):** Full 15-step completing-the-square derivation
 - **L update (Sec 4):** Explicit residual substitution, A/B identification for genomics and survival terms separately then combined
 - **F update (Sec 5):** Full quadratic grouping to A_{jk} , B_{jk} with R4 fix
 - **beta update (Sec 6):** Variance-mean decomposition $E[l^2] = \text{Var}(l) + \bar{l}^2$ shown explicitly; error-in-variables correction A_k derived (R5, R6)
 - **tau update (Sec 7):** Full $\bar{R}^2_{ij}$ expansion; explicit optimisation step (R7, R8)
 - Recompiled REVISED.pdf: **21 pages** (was 12)
2. **Ran V2.R simulation** ($n=250$, $p=1000$, $K=5$):
 - Key results: RMSE = 0.9978, C-index PCA = 0.827 vs Supervised = 0.828, PH test $p = 0.258$ (no violation), convergence in 45 iterations
3. **Wrote and rendered results/simulation_report** – 12-section comprehensive report, 23 pages via pandoc + xelatex

Deliverables committed: 5da14f3, e424d14 (pushed to origin/main)

0.8.3 Session 3 (March 10, 2026)

q(beta) derivation, modular implementation, and TDD test suite.

1. **Verified q(beta) derivation** across REVISED.tex (Sec. 6) and March 6 PDF (Sec. 6A):
 - Both agree: $A_k = \sum_i W_{ii} * E_q[l^2_{ik}]$, $B_k = \sum_i W_{ii} * z^{\{-k\}}_i * \bar{l}_{ik}$
 - Confirmed March 6 PDF Eq. 68 typo is intermediate-step only; flagged as R5 in REVISED
 - Confirmed V2.R omits $1/\sigma^2$ prior term intentionally (EBNM handles it empirically)
 - Proved z_{no_k} reuse is mathematically valid
 - Confirmed: $s_k = 1/\sqrt{A_k}$ is SMALLER when A_k is larger (not larger)
2. **Created test infrastructure** (tests/ directory):
 - tests/test_helpers.R: `assert_near`, `assert_true`, `assert_equal`, `assert_length`, etc.
 - tests/run_tests.R: master runner; exits status 1 if any tests fail

3. **TDD Red phase:** 24 failing tests first (9 groups: math identities, WLS limit, K=1 recovery, K=5 recovery, null shrinkage, error-in-variables, numerical stability, Gauss-Seidel ordering, V2 consistency)
4. **TDD Green phase:** Implemented code/update_beta.R:
 - compute_z_no_k(z, EL, EBeta, k) – partial working response
 - update_beta_k(w, z_no_k, EL_k, EL2_k, ...) – single-factor update
 - update_beta_all(w, z, EL, EL2, EBeta, ...) – full Gauss-Seidel loop
5. **Fixed 2 test logic errors** during green phase:
 - T4.1: length() returns integer (5L != 5); fixed to assert_length()
 - T6.1: s_high should be < s_low (not >); larger A_k -> smaller s_k
6. **Created derivations/qB/qBeta_update_derivation.tex** (11 pages):
 - tcolorbox styles: derivbox (gray), correctionbox (red), ebnmbox (blue), verifybox (green), codebox (orange)

Key decisions:

- Plain R test framework (no testthat/DESCRIPTION) – no external dependencies
- update_beta_k decoupled from CAVI loop; returns diagnostic fields (A, B, x, s)
- Gauss-Seidel propagation: EBeta_curr[k] updated immediately within the loop

0.8.4 Session 4 (March 11, 2026)

Demo for beta update + to-do setup for L/F/tau modules.

1. Created demos/demo_update_beta.R – 5 interactive scenarios: anatomy of A_k/B_k, signal recovery K=1, sparsity/shrinkage, error-in-variables, multi-factor K=5. Format: sep/sec/val helpers, pure narrative output (no PASS/FAIL).
2. Updated PROJECT_STATUS.md to add demos/ to the repo map.
3. Updated code/SupervisedMF_Context.md with expanded L/F/tau next-step details.

Deliverables committed: ad80c4e (pushed to origin/main)

0.8.5 Session 5 (March 12, 2026)

Full implementation of modular updates for $q(L)$, $q(F)$, and $q(\tau)$.

1. **code/update_L.R** – Three functions:

- `compute_R_k(Y, EL, EF, k)`: Partial residual matrix ($n \times p$), shared with F update
- `update_L_k(...)`: Vector EBNM – $A_L[i]$ and $B_L[i]$ are n -vectors (patient-specific Cox weights W_{ii}); two additive sources (genomics + survival)
- `update_L_all(...)`: Gauss-Seidel loop over K factors

2. **code/update_F.R** – Two functions:

- `update_F_k(...)`: Pure genomics EBNM – τ_j **cancels** in $x_j = B_F[j]/A_F[j]$ but does NOT cancel in s_j
- `update_F_all(...)`: Gauss-Seidel loop over K factors

3. **code/update_tau.R** – Three functions:

- `compute_var_term(EL, EL2, EF, EF2)`: Variance correction ($EL2 \times EF2' - EL^2 \times EF'^2$)
- `compute_expected_residual_sq(...)`: $R2_bar = \text{naive residual}^2 + \text{Var_Term}$
- `update_tau(...)`: Closed-form MLE $\tau\text{-hat}_j = n / \text{colSums}(R2_bar) + \text{ELBO proxy}$

4. **Tests** (9 groups each, all passing):

- 28 tests for `update_L.R` (T9 verifies bit-for-bit match with V2.R lines 310–321)
- 26 tests for `update_F.R` (T1.4/T1.5 verify tau cancellation numerically)
- 27 tests for `update_tau.R` (T4 verifies variance correction direction)
- Master runner: 105/105 total tests passing

5. **Demos** (5 scenarios each) and **LaTeX derivations** (compiled to PDF) for all three modules.

Key design decisions:

- `compute_R_k` shared between `update_L.R` and `update_F.R` (F sources it from `update_L.R`)
- τ does not appear in the survival term – F and τ updates are pure genomics
- `update_tau` has NO per-k loop and NO `ebnm` dependency (only base R)
- ELBO proxy computed inside `update_tau` as a convergence monitor

0.8.6 Session 6 (March 20, 2026)

Comprehensive documentation: companion docs, roxygen2, and inline comments.

1. **Created docs/ directory** with companion documentation for all four modules:

- `docs/update_beta.qmd` – Overview, math background, function reference, test + demo explanations
- `docs/update_L.qmd` – Dual-source precision/signal, `compute_R_k` shared dependency

- docs/update_F.qmd – tau cancellation property, pure genomics
 - docs/update_tau.qmd – Closed-form MLE, variance correction, ELBO proxy
2. **Created docs/Makefile:** make all renders all 4 .md files to PDF + HTML (Helvetica/Menlo fonts for Unicode Greek letter support).
 3. **Added roxygen2 package-readiness tags** to all 11 functions across 4 R scripts:
 - @export, @family (beta_update / L_update / F_update / tau_update), @seealso
 4. **Added targeted inline comments** (~3–5 per file) explaining *why* at non-obvious points.
 5. **Verification:** 105/105 tests still passing; all 8 rendered outputs (4 PDF + 4 HTML) confirmed.
-

0.8.7 Session 7 (March 20, 2026)

Modular simulation report, results/ reorganization, and Quarto conversion.

1. **Reorganized results/:**
 - Created figures/{full_sim,modular_sim}/ and tables/{full_sim,modular_sim}/
 - Moved existing V2 figures/CSVs into full_sim/ subfolders
 - Updated run_simulation.R output paths accordingly
2. **Created results/run_modular_simulation.R** (~340 lines):
 - Completely standalone – sources only the 4 modular update modules
 - calc_cox_taylor() copied verbatim from V2.R; no other V2 dependency
 - Block-by-type update ordering: all L -> all F -> all beta -> tau (vs. V2's by-factor ordering; equivalent CAVI objective)
 - Results: RMSE=0.9979, C-index=0.858, PH test p=0.44, null factor beta correctly zeroed
3. **Created results/modular_sim_report.qmd** (863 lines):
 - Self-contained Quarto PDF; CAVI runs inline with #| cache: true
 - ggplot2 figures (theme_bw), kableExtra tables, Quarto callout blocks
 - Per-figure commentary (~150–300 words each) with statistical insights
4. **Created results/simulation_report.qmd** (786 lines):
 - Sources V2.R inline with pdf(NULL) + sink() wrappers to suppress output
5. Deleted .md report files; .qmd + .pdf are the canonical report artifacts.

Deliverables committed: 9a59c3e (pushed to origin/main)

0.8.8 Session 8 (March 20, 2026)

Documentation housekeeping: CLAUDE.md, file role clarification, PROJECT_STATUS to Quarto.

1. **Created CLAUDE.md** – thin Claude Code entry point (key instructions + quick-reference table); defers to PROJECT_STATUS for full context.
2. **Discussed file role separation:**
 - README.md = GitHub front page (overview, quickstart, brief math)
 - PROJECT_STATUS.qmd/.pdf = full internal working document (session log, derivation review, algorithm design decisions, next steps)
 - CLAUDE.md = lean Claude Code instructions pointing to PROJECT_STATUS
3. **Converted PROJECT_STATUS.md to PROJECT_STATUS.qmd** for PDF rendering:
 - Added Quarto YAML header (Helvetica Neue / Menlo, TOC, numbered sections)
 - Added callout blocks for key notices (derivation errors warning, V2 recommended tip, test suite status note)
 - Deleted PROJECT_STATUS.md; .qmd + .pdf are now canonical
4. Updated MEMORY.md documentation convention to reflect the three-file structure.

Deliverables committed: pushed to origin/main

0.8.9 Session 9 (March 20–31, 2026)

Factor-wise modular CAVI: fit_modular.R, synthetic report reorganization, legacy code cleanup.

1. **Created code/fit_modular.R** (fit_supervised_mf_modular()):
 - Factor-wise Gauss-Seidel CAVI: within each iteration, cycle over $k = 1..K$ and update (L_k, F_k, β_k) in sequence before moving to $k+1$. This is the true block coordinate descent with immediate propagation — distinct from the earlier block-wise modular runner which updated all- L , then all- F , then all- β .
 - Internally sources update_L.R, update_F.R, update_beta.R, update_tau.R.
 - Returns `list(EL, EL2, EF, EF2, EBeta, EBeta2, Tau, history)`.
 - `history$converged, history$n_iter, history$rmse, history$elbo_proxy`.
2. **Created results/modular_sim_factor/run_factor_modular_simulation.R:**
 - Comprehensive analytics runner sourcing fit_modular.R.
 - Produces 8 figure pairs (PDF+PNG), 11 CSV tables for synthetic runs.
 - Designed for both synthetic and real-data modes (real-data support added fully in Session 10).
3. **Created docs/fit_modular.qmd/.pdf/.html:**

- Full code walkthrough for `fit_modular.R`, covering factor-wise loop structure, Taylor linearization, convergence criterion, and comparison with block-wise runner.

4. **Reorganized results/** into per-simulation subdirectories:

- `results/modular_sim_factor/synthetic/` — factor-wise CAVI report (canonical)
- `results/modular_sim_block/` — block-wise modular (deprecated)
- `results/full_sim/` — V2 monolithic
- `results/tables/synthetic/` and `results/figures/synthetic/` for canonical outputs

5. **Moved legacy code** to `code/legacy/`:

- `code/legacy/Supervised_Bayesian_MF.R` — V1 original
- Updated all references in `CLAUDE.md`, `README.md`, and runner scripts.

6. **Simulation results (synthetic, factor-wise, n=250, p=1000, K=5, seed=42):**

- RMSE = 0.9979 (converges to true noise SD)
- ELBO proxy: non-decreasing
- β signs: $(-, +, 0, -, +)$ — correct sign recovery, Factor 3 ($\beta=0$) zeroed
- C-index: PCA 0.827 vs Supervised ~ 0.86
- Converged at iteration 61 ($dL < 1e-3$, $dBeta < 1e-3$)

Deliverables committed: 58cda7b, dca6fe4, 110872c, 0b560db, 94ed9a2

0.8.10 Session 10 (March 31, 2026)

Real PDAC data integration: 7 cohorts, pooled RNA-seq, test suite, and PDAC report.

0.8.10.1 10.1 — Runner extension for real data

Extended `results/modular_sim_factor/run_factor_modular_simulation.R` with:

Configuration block (env-var overrides for scripted/Longleaf use):

```
data_mode      <- "synthetic"    # "synthetic" or "real"
dataset_name   <- "TCGA_PAAD"    # single-dataset mode
run_all        <- FALSE          # TRUE: loop all 7 PDAC cohorts
top_n_genes    <- 5000           # top-N by variance; NULL = all
K              <- 5
pdac_data_root <- Sys.getenv("PDAC_DATA_ROOT", unset = ...)
```

New helpers: - `filter_top_genes(Y, gene_names, top_n)` — selects top-N most-variable columns; passthrough if $top_n \geq p$ or $top_n = \text{NULL}$. Fixed with `drop = FALSE` to prevent silent matrix-to-vector coercion when $top_n = 1$. - `load_real_data(dataset_name, pdac_root, top_n)` — loads via `load_data_internal.R` (symlink trick to handle relative data/original/paths), subsets to `keep == 1` samples, transposes ex matrix, extracts gene symbols, applies

variance filter, **column-centres Y**. - `pool_datasets(ds_list)` — intersects gene names, aligns columns, row-binds Y, concatenates time/status for horizontal integration.

Multi-dataset loop (when `run_all = TRUE`): - Iterates over all 7 PDAC cohorts; accumulates per-dataset summary rows. - Writes `results/tables/PDAC_cross_dataset/cross_dataset_summary.csv`.
 - After loop: pools TCGA_PAAD + Dijk + PACA_AU_seq → fits one model → outputs to `results/tables/PDAC_pooled_rnaseq/` and `results/figures/PDAC_pooled_rnaseq/`.

0.8.10.2 10.2 — Critical bug fixes

Three bugs identified and fixed:

Bug	Root cause	Fix
RMSE increasing on real data	Missing column centring	<code>Y <- sweep(Y, 2, colMeans(Y), "-")</code> in <code>load_real_data()</code>
Dataset column wrong in summary	<code>run_pipeline()</code> used global <code>dataset_name</code>	<code>row\$Dataset <- ds</code> after <code>run_pipeline()</code> in loop
Filter crash when <code>top_n = 1</code>	<code>Y[, scalar_idx]</code> drops matrix to vector	<code>Y[, keep_idx, drop = FALSE]</code>

Column centring is the most important fix. Without it, SVD initialisation captures the global mean expression in Factor 1; CAVI then fights the mean artefact each iteration, causing RMSE to *increase* rather than decrease. Column centring is standard preprocessing for matrix factorisation on genomics data.

0.8.10.3 10.3 — Convergence results (K=5, top_5000 genes, max_iter=300)

Dataset	Platform	n	Converged	Iterations	Final RMSE
TCGA_PAAD	RNA-seq	144	TRUE	14	~0.99
Dijk	RNA-seq	90	TRUE	~20	~0.99
PACA_AU_seq	RNA-seq	52	TRUE	~30	~0.99
Moffitt_GEO_array	Microarray	123	TRUE	~50	~0.99
PACA_AU_array	Microarray	63	TRUE	~60	~0.99
Puleo_array	Microarray	288	TRUE	~80	~0.99
CPTAC	Proteomics	129	TRUE	259	~0.99

CPTAC required `max_iter = 300` (was 100, then 250); it converged at iteration 259.

0.8.10.4 10.4 — Test suite: `tests/test_real_data_loading.R`

77 tests, auto-skip if PDAC_DATA_ROOT not found (CI-safe): - **T1** (6 tests): `filter_top_genes()` — variance selection, passthrough, NULL, single-gene (`drop=FALSE`), dimension consistency - **T2** (63 tests = 7×9): all 7 datasets load without error, correct dimensions, no NAs, time > 0, status $\in \{0,1\}$, numeric matrix, row/col alignment, Y is column-centred (`colMeans` ~ 0 to tolerance $1e-10$), `gene_names` length matches p - **T3** (6 tests): `pool_datasets()` — non-empty intersection, `nrow == sum(n)`, `ncol == intersection length`, no NAs, `dataset_labels` length, error on empty intersection - **T4** (2 tests): synthetic regression — `RMSE` $\in [0.95, 1.05]$, `convergence == TRUE`

77/77 passing; 105/105 core module tests still passing.

0.8.10.5 10.5 — PDAC report

`results/modular_sim_factor/PDAC/factor_modular_sim_report_PDAC.qmd` — fully standalone Quarto report rendering to PDF + HTML.

Self-contained model background: - Model equations ($Y = LF' + E$, Cox PH), EBNM prior, CAVI pseudocode, fit settings table ($K=5$, `max_iter=300`, `tol=1e-3`, `top_5000` genes)

“How to Read These Results” section: - Explains all 8 figure types (RMSE trace, ELBO proxy, β estimates, GEP heatmap, KM curves, PH test, C-index comparison, τ distribution) with what/how/why narrative — no need to reference the synthetic report.

Per-dataset sections (7 datasets + pooled fit): - Convergence history table (first 5 + last 5 iterations), β estimates, factor summary table, C-index comparison, PH test, top-10 genes per GEP, τ distribution figure (fig8)

Dataset-specific figure commentary — computed R values emit short italic paragraphs after each figure, e.g.: - RMSE: drop magnitude and iteration count - ELBO: ascending vs non-monotone check - β : count of prognostic vs null factors - Factor summary: Cox β vs log-rank p consistency - GEP heatmap: dominant factor PVE and prognostic status - KM: which factors show significant stratification - PH test: pass/fail with action language - C-index: contextual (supervised wins/comparable/lower vs PCA) - τ : platform-specific interpretation (RNA-seq/Microarray/Proteomics)

Cross-dataset sections: - Executive summary table (all 7 datasets: convergence, RMSE, C-index) - Cross-platform PACA-AU pair comparison (β correlation, top-gene Jaccard) - Within-modality comparisons (RNA-seq trio, microarray trio) - CPTAC proteomics standalone section - Pooled RNA-seq section (dataset membership, pooled fit results, batch caveat)

`ds_display()` helper replaces underscores with spaces in all LaTeX caption strings (prevents “Missing \$ inserted” from TeX math-mode interpretation).

Renders to PDF (xelatex, via Quarto) and HTML without errors.

0.8.10.6 10.6 — Future directions recorded

- **Prediction task using hold-out test set:** Apply trained model to a held-out subset of each cohort; evaluate C-index on out-of-sample patients. Requires a `predict_supervised_mf()` function to propagate new patients through the learned F and β .
- **K selection via cross-validation:** Held-out log-likelihood or C-index as objective; compute-intensive, planned for Longleaf HPC.
- **Full ELBO tracking:** Currently a genomics-only proxy; survival Taylor term and KL divergences not yet included.
- **Batch-corrected pooled model:** Apply ComBat or similar before pooling RNA-seq cohorts to remove dataset-of-origin effects.

Deliverables committed: b3e23b5, pushed to origin/main as part of session 11.

0.8.11 Session 11 (March 31, 2026)

Phases 1–7: Batch correction, multi-init, prior threading, hold-out prediction, feature selection, K selection, and report.

0.8.11.1 11.1 — Phase 1: limma batch correction for pooled RNA-seq

Added `apply_batch_correction(Y, batch_labels)` helper to `run_factor_modular_simulation.R`. Uses `limma::removeBatchEffect()` applied after `pool_datasets()` for the pooled RNA-seq fit. `limma` is a Bioconductor package (installed via `BiocManager::install("limma")`). The function transposes Y before calling `limma` (which expects genes $\times$ samples) and transposes back. Controlled by `batch_correct = TRUE` (default) / `BATCH_CORRECT` env var.

0.8.11.2 11.2 — Phases 2 + 3: Multi-init and prior_family threading

Multi-init (code/fit_modular.R): - New parameters: `prior_family` and `init_method` added to `fit_supervised_mf_modular()`. - `init_method = "random"` initialises L and F with `rnorm(sd = 0.1 * sd(Y))` (scaled to prevent Cox Taylor overflow at startup). - Multi-init loop in runner: when `n_init > 1` and `init_method = "random"`, runs `n_init` independent restarts and keeps the fit with the highest final ELBO proxy. Results saved to `multi_init_elbos.csv`.

Prior threading: - `prior_family` is now passed through to all three EBNM update calls inside the factor loop: `update_L_k(..., prior_family)`, `update_F_k(..., prior_family)`, `update_beta_k(..., prior_family)`. Previously the argument existed in the signature but was not forwarded (defaulted to `"point_normal"` everywhere).

0.8.11.3 11.3 — Phases 4–6: Hold-out prediction, feature selection, K selection

code/predict.R — predict_supervised_mf(Y_test, EF, EBeta, lambda): - Pseudo-inverse projection: $\hat{L}_{\text{test}} = Y_{\text{test}} \hat{F} (\hat{F}^T \hat{F} + \lambda I)^{-1}$ - Risk score: $\hat{L}_{\text{test}} \hat{\beta}$ - lambda = 1e-8 ridge regularisation to handle rank-deficient F.

code/train_test_split.R — stratified_split(status, test_frac, seed): - Splits events and censored patients separately, each at test_frac fraction. - Guarantees ≥ 1 event and ≥ 1 censored in both train and test. - Returns list(train_idx, test_idx, n_train, n_test).

code/feature_selection.R — cox_feature_selection(Y, time, status, p_thresh): - Fits one univariate Cox model per gene; returns indices of genes with $p < p_thresh$. Applied to training data only to prevent survival leakage.

code/select_K.R — auto_prune_K() + select_K_cv() stub: - auto_prune_K(Y, time, status, K_max, beta_thresh, pve_thresh, ...): fits with K_max, flags factor k as active if $|\beta_k| > \text{beta_thresh}$ OR $\text{PVE}_k > \text{pve_thresh}$. Returns K_effective and the full fit object. - select_K_cv(): errors by design with a message pointing to Longleaf HPC.

tests/test_predict.R: 19 tests for predict_supervised_mf() and stratified_split(). Total test count: 105 + 19 = **124/124 passing**.

0.8.11.4 11.4 — Phase 7: Report update, rendering, documentation

PDAC report updated with: - Hold-out C-index section (PN K=5, stratified 80/20, 7 cohorts) - Conclusion table updated with feature status

Runner updated with env-var overrides for all new parameters: PRIOR_FAMILY, N_INIT, INIT_METHOD, BATCH_CORRECT, HOLDOUT_EVAL, FEATURE_SELECTION, K_SELECT.

Hold-out results (PN K=5, 80/20 split, seed=42):

Dataset	C Train	C Test (SBMF)	C Test (PCA)
TCGA_PAAD	0.385	0.341	0.327
CPTAC	0.315	0.389	0.371
Dijk	0.383	0.381	0.424
Moffitt_GEO_array	0.429	0.475	0.469
PACA_AU_array	0.353	0.405	0.262
PACA_AU_seq	0.210	0.185	0.178
Puleo_array	0.340	0.396	0.374

Deliverables committed: 89ff1ff, ff1a121, 3262821

0.8.12 Session 12 (April 1, 2026)

Prior family comparison (executed), K auto-prune (executed on all 7 cohorts), PDAC report editorial review.

0.8.12.1 12.1 — Prior family comparison: `run_prior_k_comparison.R`

New script: `results/modular_sim_factor/run_prior_k_comparison.R`. Standalone (no global-variable dependencies on main runner). For each of the 7 datasets:

1. Reads existing PN K=5 holdout results from CSVs (avoids redundant refit).
2. Fits `point_laplace` K=5 with 80/20 hold-out; saves to `{dataset}_pl/`.
3. Runs `auto_prune_K(K_max=10)` to identify `K_effective`; saves PVE table and scree figure to `{dataset}/k_selection_pve.csv` and `figK_pve_scree.pdf/png`.
4. Refits with K=`K_effective` (PN) + hold-out; saves to `{dataset}_Keff/`.
5. Accumulates all results into two cross-dataset summary CSVs.

0.8.12.2 12.2 — Results: prior comparison

`results/tables/PDAC_cross_dataset/prior_comparison.csv`

`point_laplace` K=5 vs. `point_normal` K=5 (hold-out C-index):

Dataset	C Test PN K=5	C Test PL K=5	ΔC
TCGA_PAAD	0.341	0.707*	+0.366
CPTAC	0.389	0.605	+0.216
Dijk	0.381	0.483	+0.102
Moffitt_GEO_array	0.475	0.580*	+0.105
PACA_AU_array	0.405	0.667	+0.262
PACA_AU_seq	0.185	0.889	+0.704
Puleo_array	0.396	0.594	+0.198

* = hit 300-iteration convergence limit. `point_laplace` outperforms `point_normal` in all 7 cohorts. This is the primary motivating finding for switching the default prior to `point_laplace` in future analyses.

0.8.12.3 12.3 — Results: K auto-prune

`results/tables/PDAC_cross_dataset/k_selection_summary.csv`

Dataset	K_max	K_effective
TCGA_PAAD	10	9
CPTAC	10	8
Dijk	10	10 (saturated)
Moffitt_GEO_array	10	10 (saturated)

Dataset	K_max	K_effective
PACA_AU_array	10	8
PACA_AU_seq	10	9
Puleo_array	10	10 (saturated)

K=5 was insufficient for every dataset. Three cohorts saturate at K_max=10, suggesting their true dimensionality exceeds 10. Re-running with K_max=15 on Longleaf HPC is recommended for these three.

PN K_eff vs. PN K=5 hold-out C-index improvement was positive for all cohorts (largest: TCGA_PAAD +0.28, PACA_AU_array +0.33, PACA_AU_seq +0.56).

0.8.12.4 12.4 — PDAC report editorial review

results/modular_sim_factor/PDAC/factor_modular_sim_report_PDAC.qmd overhauled:

- **Executive Summary:** Expanded callout to cover all four key findings (convergence, in-sample results, prior comparison, K selection).
- **Fit Settings table:** Added notes that K is now data-driven and prior family is compared in later sections.
- **CPTAC section:** Removed duplicated factor summary table; replaced with concise proteomics spotlight with cross-references to new sections.
- **Prior Comparison section:** Fixed broken sentence fragment; quantified top gains; noted the missing PL + K_eff experiment.
- **K Selection section:** Moved CV runtime note after the tables; interpretation now explicitly names the three saturating cohorts.
- **Conclusion:** Completely rewritten as a narrative covering clinical motivation, what SBMF does, four key findings, six current limitations, and six future directions.

Deliverables committed and pushed: 49dd8f6, ab23228

0.8.12.5 12.5 — Open items / next session priorities

1. ~~Run point_laplace at K_effective per dataset (the expected best configuration).~~ **Done in Session 13.**
2. Gene set enrichment analysis on top-loaded genes per GEP (fgsea / Enrichr).
3. K selection via CV on Longleaf HPC (select_K_cv() stub is ready).
4. Re-run Dijk, Moffitt, Puleo with K_max=15 to probe true dimensionality.
5. ~~Full ELBO tracking (add Cox likelihood and KL terms to ELBO proxy).~~ **Done in Session 13.**

0.8.13 Session 13 (April 1–2, 2026)

Full ELBO tracking implemented; point_laplace at K_eff run for all 7 cohorts; all results re-run with consistent methodology; PDAC report updated.

0.8.13.1 13.1 — Full ELBO implementation

New file: `code/compute_elbo.R`. Provides two functions:

- **`compute_ebnm_kl(ebnm_log_lik, A, x, mean_q, second_q)`** — extracts the KL divergence $E_q[\log g(\theta)] - E_q[\log q(\theta)]$ from one EBNM update by subtracting the expected pseudo-log-likelihood from the returned marginal log-likelihood. Works for vectors (L, F updates) and scalars (β update). Returns a scalar ≤ 0 .
- **`compute_survival_elbo(logPL, w, EL, EL2, EBeta, EBeta2)`** — computes $E_q[\log \text{PL}(t, \delta \mid L, \beta)]$ via second-order Taylor approximation: $\log \text{PL}(\eta_0) - \frac{1}{2} \sum_i w_i \cdot \text{Var}_q(\eta_i)$, where $\text{Var}_q(\eta_i) = \sum_k [EL2_{ik} \cdot EBeta2_k - EL_{ik}^2 \cdot EBeta_k^2]$.

`code/fit_modular.R` updated (7 edits): - `calc_cox_taylor()`: returns $\$logPL$ (Breslow partial log-likelihood at η_0) - `source("code/compute_elbo.R")` added alongside other sources - `history$elbo_full` initialized and populated each iteration - KL accumulators (`kl_L`, `kl_F`, `kl_beta`) filled inside the k-loop from EBNM outputs - Full ELBO = `elbo_proxy` + `surv_elbo` + `sum(kl_L)` + `sum(kl_F)` + `sum(kl_beta)` - Verbose print shows both ELBOs; convergence trim extended to `elbo_full`

`tests/test_elbo.R` (new, 15 tests):

- Section 1 (4 tests): `calc_cox_taylor` backward compat + new $\$logPL$ field
- Section 2 (3 tests): `compute_ebnm_kl` — scalar output, scalar inputs, $KL \leq 0$
- Section 3 (3 tests): `compute_survival_elbo` — scalar output, zero-variance = $\log PL$ exactly, uncertainty reduces value
- Section 4 (5 tests): integration via `fit_supervised_mf_modular` — `elbo_full` present, finite, all-iters tracking

Rscript `tests/run_tests.R`: **139/139 passing** (24 β + 28 L + 26 F + 27 τ + 19 predict + 15 elbo).

0.8.13.2 13.2 — Complete 4-condition prior comparison

`run_prior_k_comparison.R` updated to run all 4 conditions for all 7 datasets:

1. `point_normal K=5 (PN_K5)`
2. `point_laplace K=5 (PL_K5)`
3. `point_normal K_eff (PN_Keff)`
4. `point_laplace K_eff (PL_Keff)` — **new**

All conditions re-run with consistent methodology (elbo_full as ELBO_Final). New {dataset}_pl_Keff/ directories created for all 7 datasets with factor_summary_table.csv, holdout_cindex.csv, and convergence_history.csv (ELBO_Proxy + ELBO_Full columns).

Key result — PL K_eff vs. PN K_eff:

Dataset	PN K_eff C_Test	PL K_eff C_Test	Winner
TCGA_PAAD	0.625	0.620	PN
CPTAC	0.635	0.623	PN
Dijk	0.544	0.510	PN
Moffitt_GEO_array	0.530	0.550	PL
PACA_AU_array	0.738	0.738	Tie
PACA_AU_seq	0.741	0.778	PL
Puleo_array	0.674	0.649	PN

Conclusion: The dominant driver of improvement is K selection (K_eff vs K=5), not prior family. At K_eff, point_normal wins in 4/7 cohorts, point_laplace wins in 2/7, tie in 1/7. The advantage of point_laplace is strongest when K is fixed at 5.

0.8.13.3 13.3 — PDAC report updates

- Fixed \$\$ LaTeX artifact in Key Findings (was rendering as literal \$+\$0.012)
- Prior Family section: updated intro to mention 4 conditions; table extended to 4 conditions with new $\delta(\text{PL K_eff} - \text{PN K}=5)$ column; plot extended with 4th condition (purple diamond); interpretation updated with actual PL K_eff results
- New “Full ELBO vs. Proxy” subsection with convergence comparison plot (TCGA_PAAD)
- Key Findings “Prior family”: updated to reflect mixed PL K_eff result
- Limitation 6: replaced “prior comparison incomplete” with “ELBO not yet used for model selection”
- Future Directions: removed completed items (PL K_eff, full ELBO); renumbered remaining
- Re-rendered: both PDF and HTML regenerated successfully

0.8.13.4 13.4 — Open items / next session priorities

1. Gene set enrichment analysis on top-loaded genes per GEP (fgsea / Enrichr).
2. K selection via CV on Longleaf HPC (select_K_cv() stub is ready).
3. Re-run Dijk, Moffitt, Puleo with K_max=15 to probe true dimensionality.
4. ELBO-based model selection: use elbo_full as criterion for K selection and prior comparison.

0.8.14 Session 14 (April 5–8, 2026)

Lab meeting preparation session.

0.8.14.1 14.1 — Slide deck creation and revision

1. Created V1 slide deck (presentation/walther_lab_meeting_04_09_2026/V1/lab_meeting_april9) — initial draft covering model motivation, algorithm, synthetic validation, and PDAC results.
2. Created V2 slide deck with revised slide text, tighter layout, and updated PDAC figures (presentation/walther_lab_meeting_04_09_2026/V2/lab_meeting_april9_V2.qmd).
3. Added dataset description table on the PDAC cohorts slide to orient the audience.

0.8.14.2 14.2 — Slide-text corrections

Fixed 4 factual discrepancies identified during review:

1. CPTAC iteration count — corrected to match the value in the convergence table.
2. RMSE direction — clarified that lower RMSE is better; phrasing was ambiguous.
3. Puleo factor labels — aligned slide labels with the factor summary table output.
4. TCGA KM language — softened “significant” to match actual PH test p-value.

0.8.14.3 14.3 — Companion document and repository hygiene

1. Created companion LaTeX document (presentation/walther_lab_meeting_04_09_2026/Notes/lab) — 21-page prep guide covering figure analysis, Q&A preparation, and review sections for each slide.
 2. Fixed .gitignore: removed the `**/*.tex` rule that was silently blocking companion.tex from being tracked; added `**/*.out` to suppress LaTeX auxiliary output files.
 3. Reorganized the presentation directory into V1/V2/Notes subdirectories for cleaner version management.
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0.8.15 Session 15 (April 9, 2026)

Post-lab-meeting fixes and feedback processing.

0.8.15.1 15.1 — Lab meeting presentation

1. Presented at Rashid lab meeting; received structured feedback from Naima Rashid and Tyler Humphreys covering model clarity, C-index computation, code organization, and research directions.

0.8.15.2 15.2 — Figure 3 permutation bug fix

1. Fixed a non-bijective permutation bug in Figure 3 (synthetic $\hat{\beta}$ comparison): `apply(abs(cors), 2, which.max)` was used to align estimated factors to true factors, but this can assign the same true factor to multiple estimated factors when two estimated factors are most correlated with the same true factor.
2. Replaced with a `bijective_match()` helper implementing greedy global-maximum assignment: iteratively selects the (estimated, true) pair with the largest absolute correlation, assigns it, removes both from consideration, and repeats — guaranteeing a one-to-one mapping.

0.8.15.3 15.3 — Synthetic simulation seed fix

1. Changed synthetic simulation seed from 42 to 222: seed 42 produced a data configuration where True F1 and True F5 (the null factor) were mixed after permutation alignment, causing the model to assign $\hat{\beta}_{\text{null}} = -0.77$ to the null factor — an artifact of the seed, not the model.
2. Seed 222 cleanly separates all 5 factors and correctly zeroes the null factor ($\hat{\beta} \approx 0$), giving a cleaner validation of the supervised component.

0.8.15.4 15.4 — Companion document update

1. Updated companion doc V2 (`presentation/walther_lab_meeting_04_09_2026/Notes/lab_meeting_04_09_2026`) to reflect the corrected Figure 3 permutation alignment and seed change.

0.8.15.5 15.5 — Open items / next session priorities

1. V3 slide deck — targeted clarifications on slides covering model specification, objective function, EBNM, CAVI, and synthetic validation, in response to lab meeting feedback.
2. C-index code fix — use the model's own β (i.e., `EL %*% EBeta`) to form the linear predictor instead of refitting a new `coxph` on training data; this tests the model's survival predictions rather than a post-hoc re-fitted Cox model.
3. Create `DECISIONS.md` — 12 architectural decision entries documenting key modelling and implementation choices made between Feb 12 and Apr 9 (e.g., Gauss-Seidel ordering, orthogonalization off, Taylor approximation for survival ELBO).
4. Create `ROADMAP.md` — research directions: incorporating F'Y term in survival model (R1), adding λ scaling parameter (R2), cross-cohort validation (R3), inter-dataset normalization strategy (R4), K selection strategy (R5), `{targets}` workflow for reproducibility (R6).
5. Create `config/globals.yml` — centralize hard-coded parameters (convergence thresholds, `K_max` defaults, simulation seeds) currently scattered across `fit_modular.R`, `select_K.R`, and the simulation runner.
6. Repo architecture review — propose a cleaner directory layout without moving files yet; surface ambiguities in current structure for discussion.
7. Re-render synthetic and PDAC simulation reports after the C-index fix to update all reported C-index values.